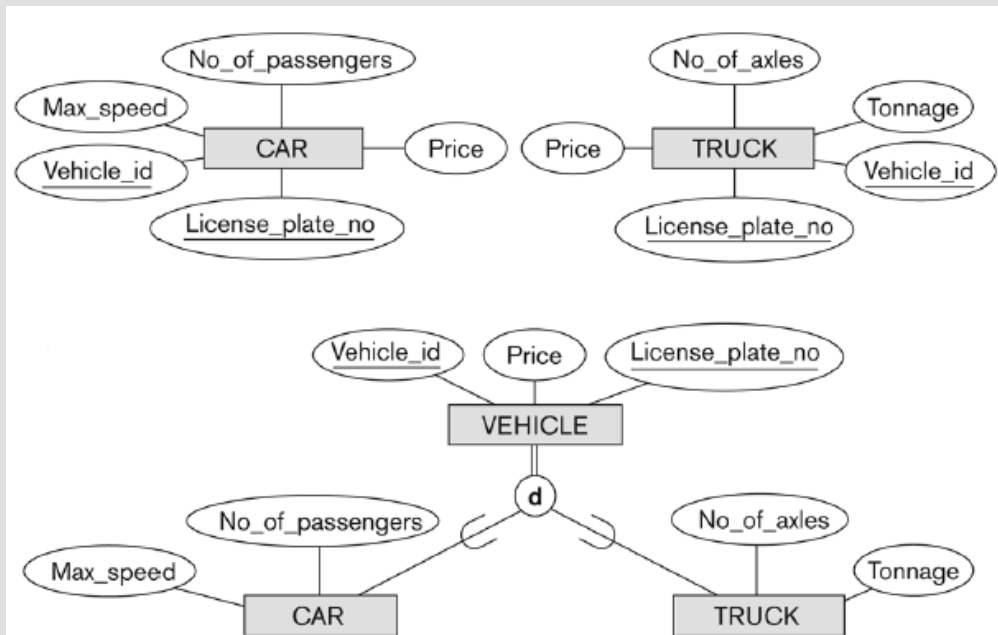


Lecture 4: The Enhanced Entity-Relationship (EER) Model

CS2300: File Structures and Introduction to Database Systems

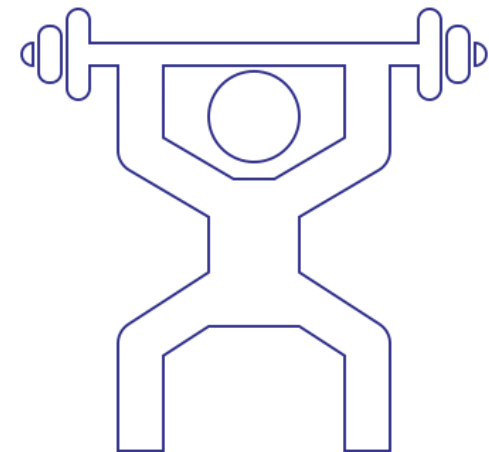
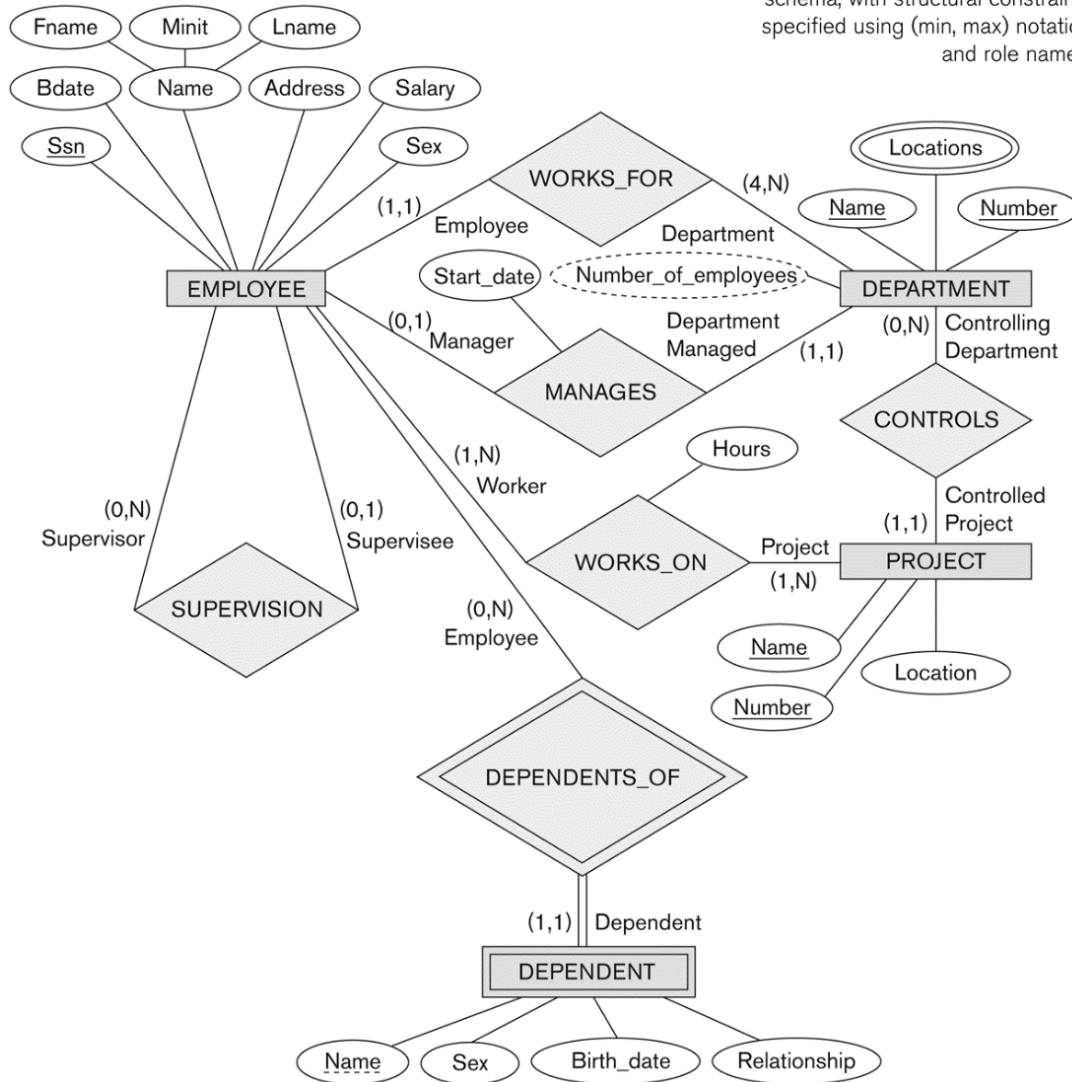
San Yeung



Generalization, (a) Two entity types, CAR and TRUCK
(b) Generalizing CAR and TRUCK into the super VEHICLE

Figure 3.15

ER diagrams for the company schema, with structural constraints specified using (min, max) notation and role names.



The Enhanced Entity-Relationship (EER) Model

- EER = Enhanced ER or Extended ER
- EER Model Concepts
 - Includes all modeling concepts of basic ER
 - Additional concepts:
 - subclasses/superclasses
 - categories (UNION types)
 - specialization/generalization
 - attribute and relationship inheritance
- The additional EER concepts are used to model applications more completely and more accurately

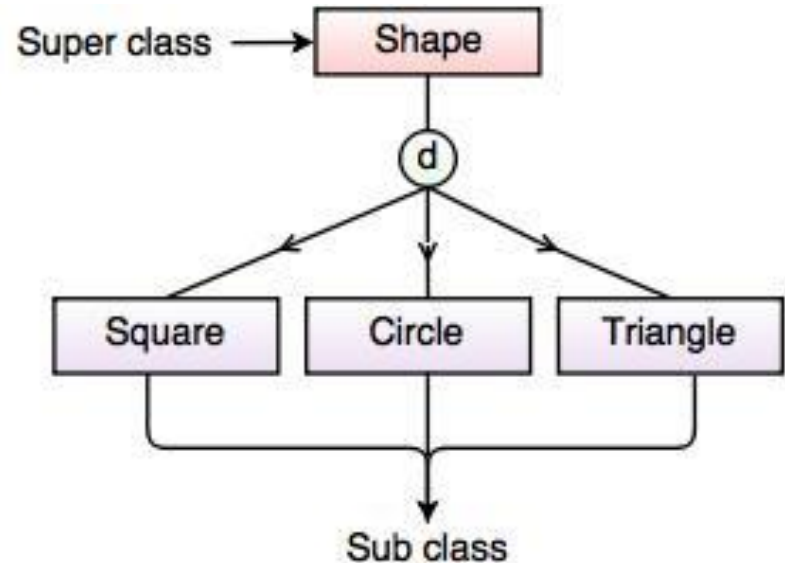
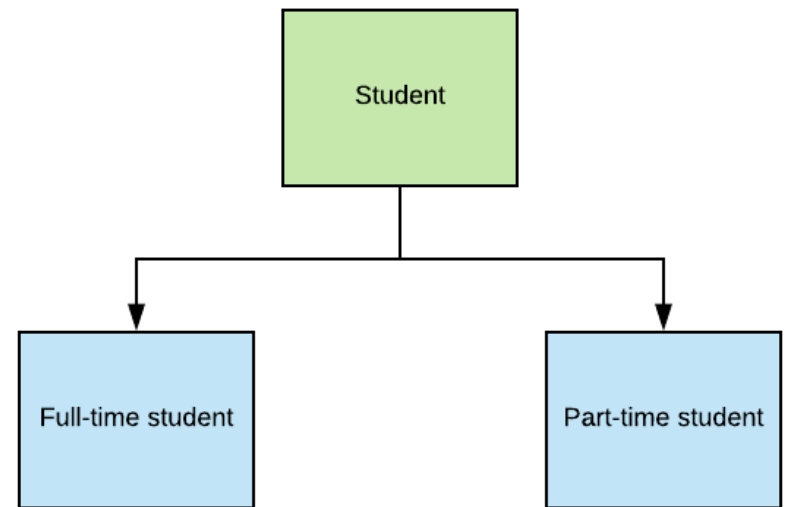


Fig. Super class/Sub class Relationship

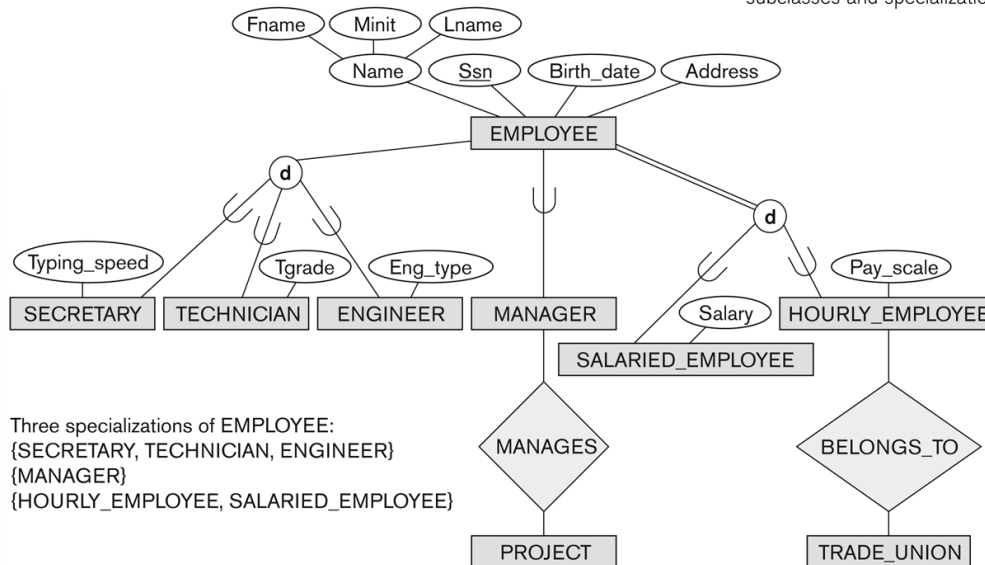
Subclasses and Superclasses

- An entity type may have additional meaningful subgroupings of its entities
- EER diagrams extend ER diagrams to represent these additional subgroupings, called *subclasses* or *subtypes*



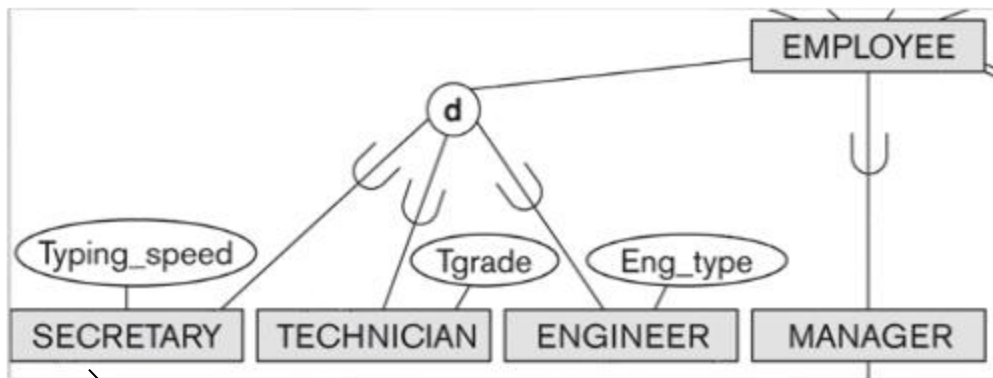
Subclasses and Superclasses

Figure 4.1
EER diagram notation to represent
subclasses and specialization.



- *How many subclasses are in here?*
- *Can an employee belong to more than one subclass?*

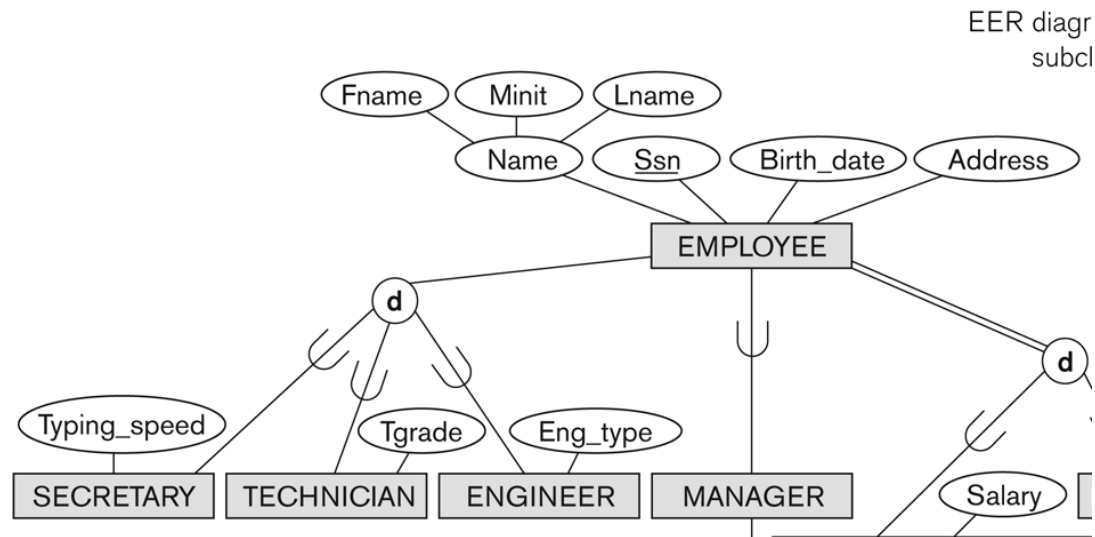
Subclasses & Superclasses



local subclass attributes

- “superclass/subclass” relationships:
 - EMPLOYEE/SECRETARY
 - EMPLOYEE/TECHNICIAN
 - EMPLOYEE/MANAGER
- also called **IS-A** relationships
 - SECRETARY IS-A EMPLOYEE,
 - TECHNICIAN IS-A EMPLOYEE
 -

Question



What would be the attribute set for the entity 'SECRETARY'?

- a) {typing_speed}
- b) {typing_speed, name, ssn, birthdate, address}
- c) {name, ssn, birthdate, address}



Come claim your inheritance!



Attribute Inheritance

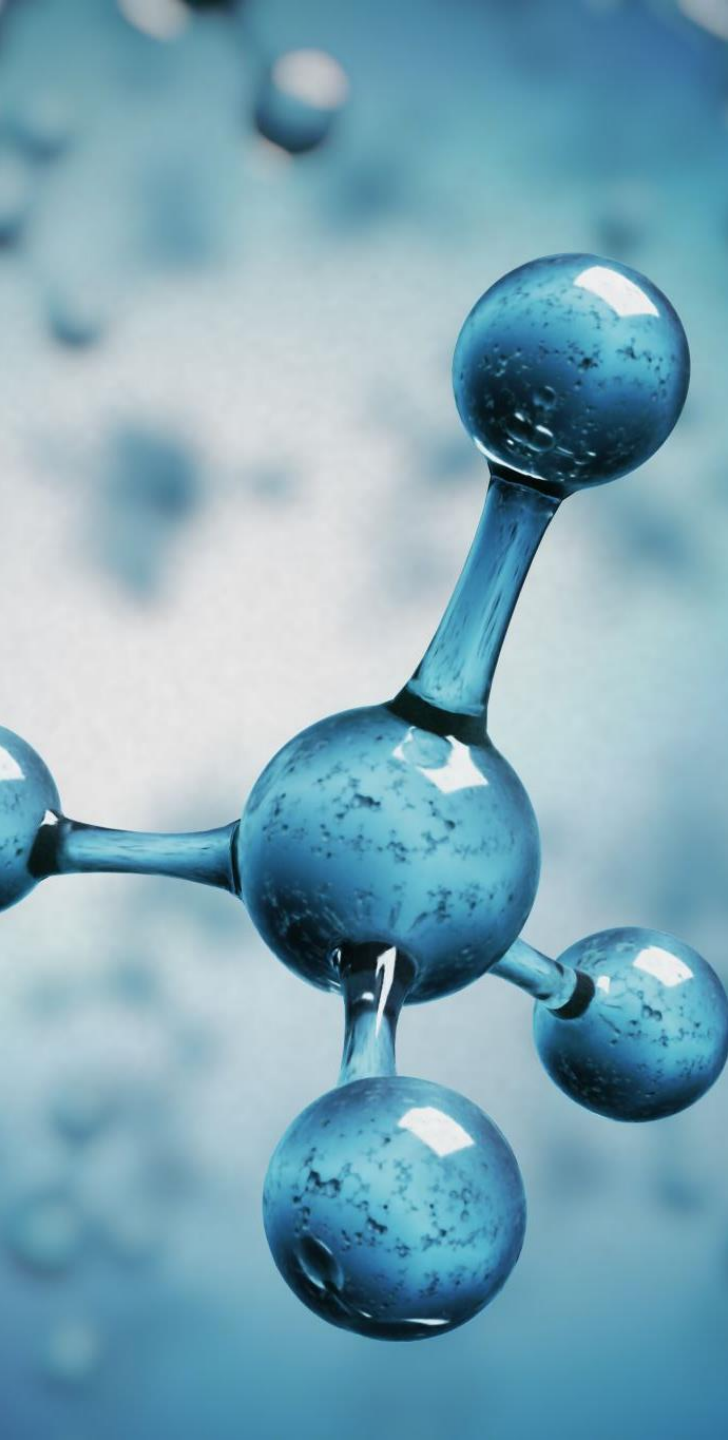
- An entity that is member of a subclass *inherits*
 - **All** attributes of the entity as a member of the superclass
 - **All** relationships of the entity as a member of the superclass

Subclasses & Superclasses

The subclass member is the same entity in a ***distinct specific role***.

An entity **cannot** exist in the database merely by being a member of a subclass.

It is **not necessary** that every entity in a superclass be a member of some subclass.

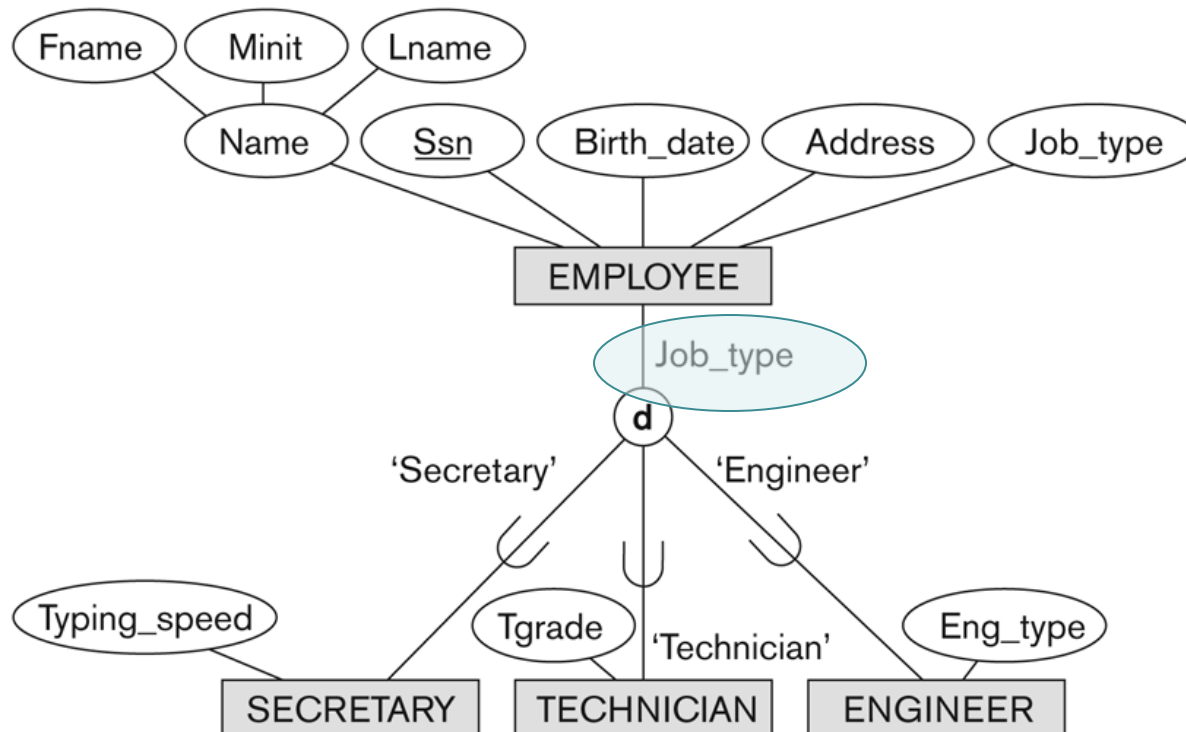


Specialization and Generalization

- EER diagrams allow refinements within the structures of entity types
- **Specialization:**
 - top-down refinement into **superclasses** and **subclasses**
- **Generalization:**
 - groups entity types
 - bottom up synthesis

Specialization

- The set of subclasses is based upon some ***distinguishing characteristics*** of the entities in the superclass



Types of Specialization

Attribute-Defined Subclasses

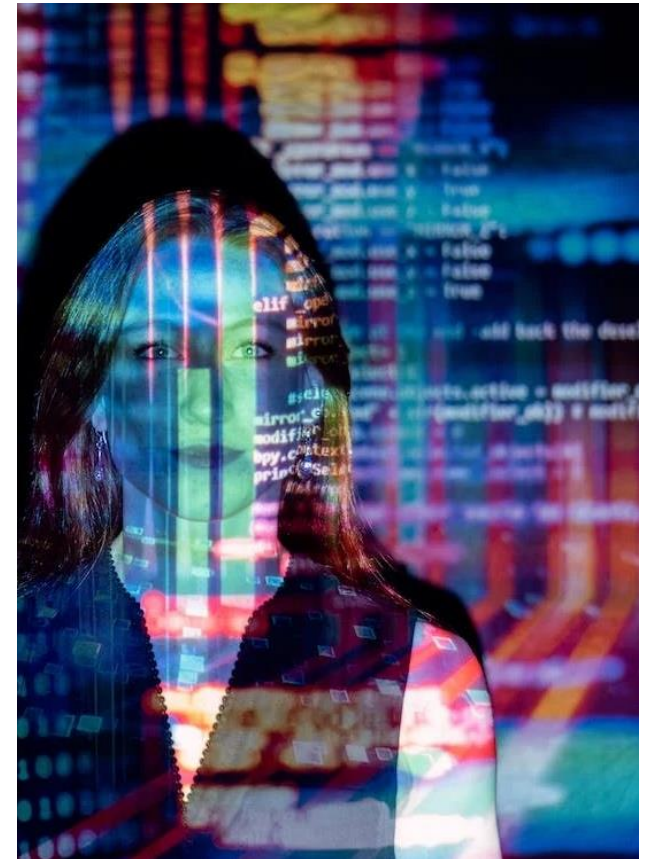
- Defined by a shared superclass attribute
- Example: the **job_type** attribute of an “EMPLOYEE” superclass

Predicate-Defined Subclasses

- Defined by conditions on superclass attributes.
- Example: the “EMPLOYEE” superclass can define a subclass that includes only high earners (e.g., **salary** > 100K).

User-Defined Subclasses

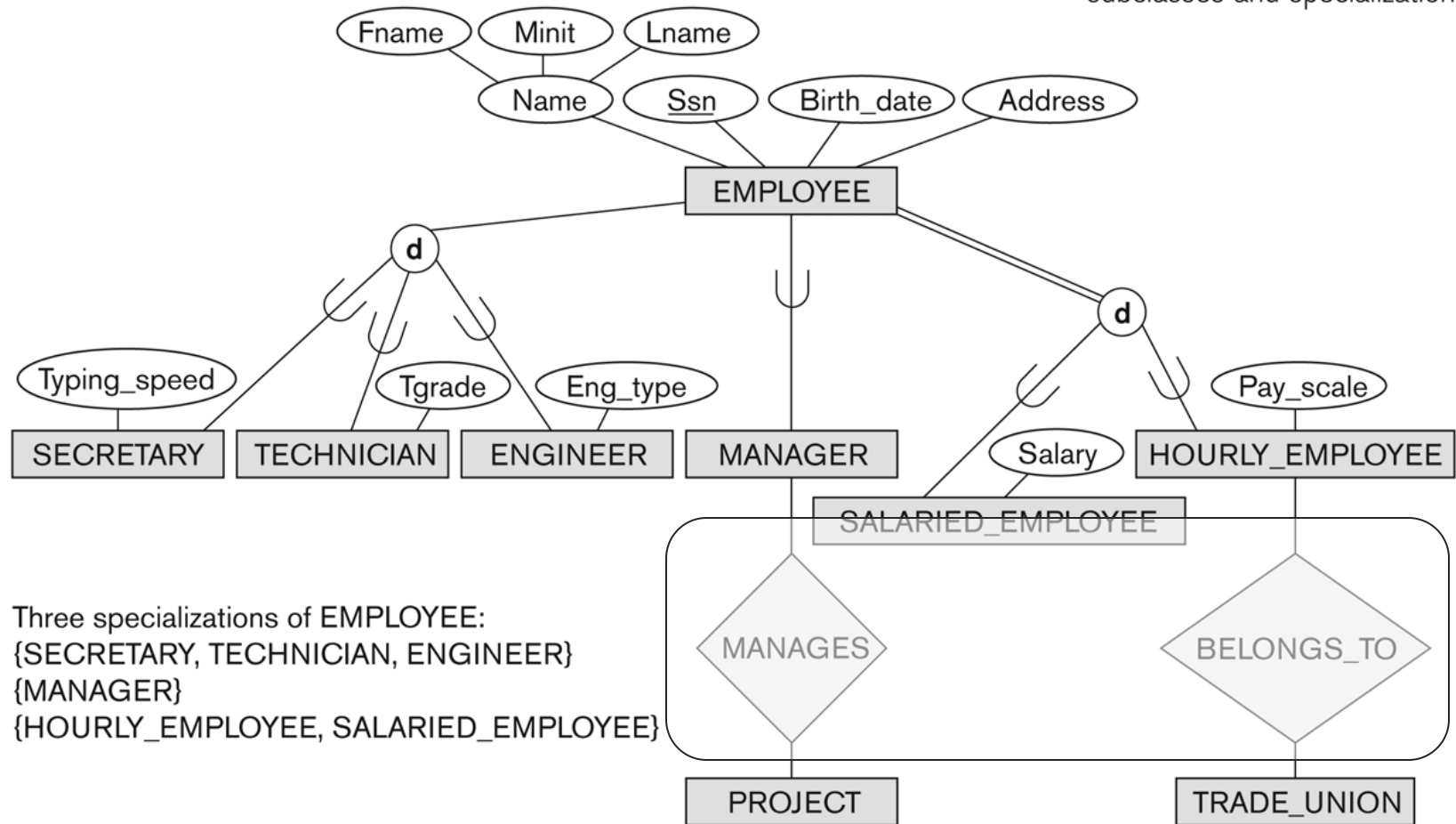
- Membership is determined based on **external criteria** at the time of data insertion.
- Example: a user might place a new employee entry into a “Temporary Staff” subclass regardless of other attribute values.



Specialization

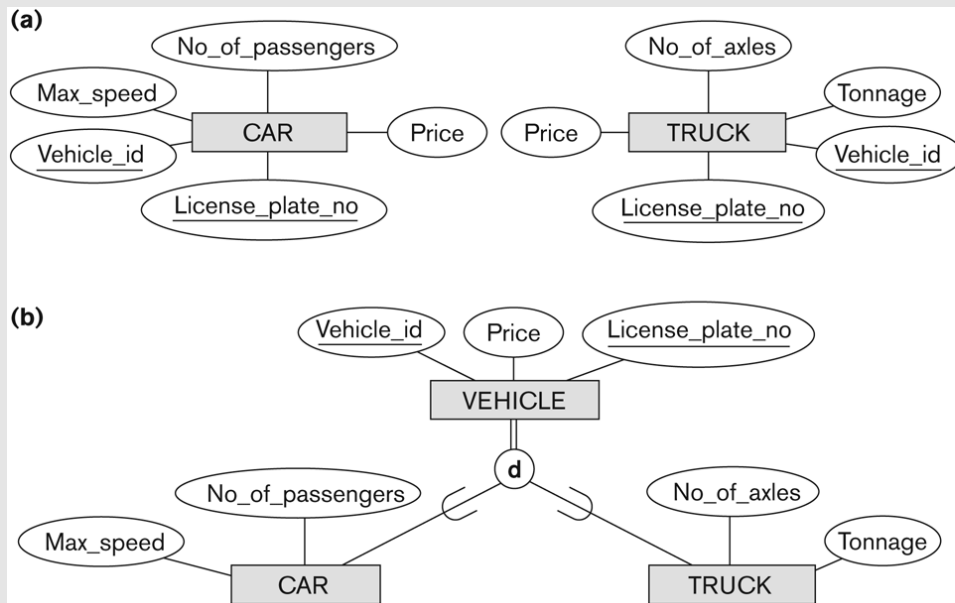
Figure 4.1

EER diagram notation to represent subclasses and specialization.



Subclasses can also participate in specific relationship types

Generalization



- Generalization is the **reverse** of the specialization process
- Original classes (with common features) become its subclasses

From Subclass to Weak Entity Type

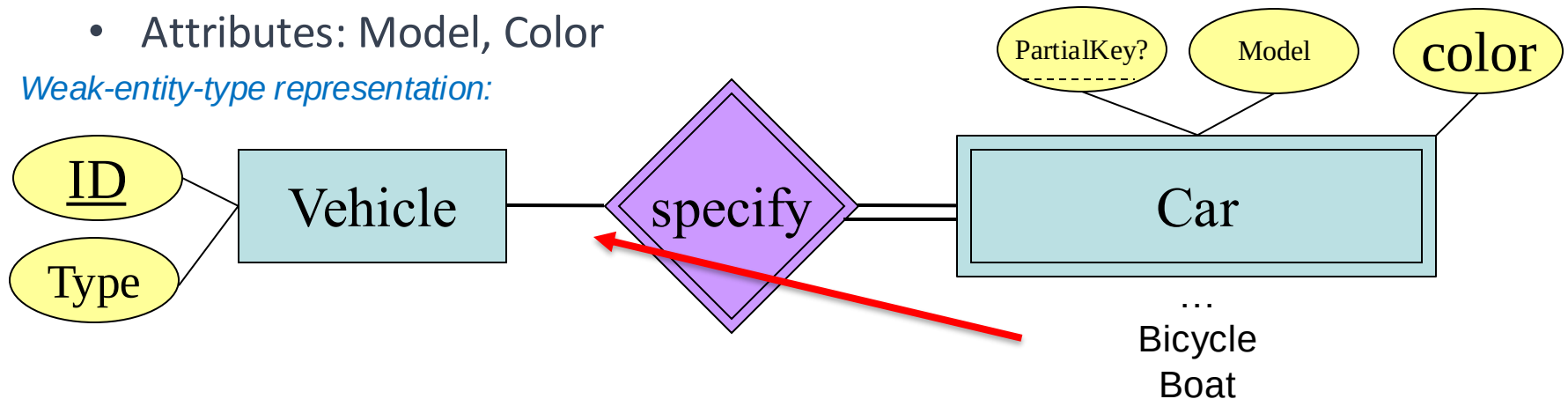
Superclass: Vehicle

- Attributes: VehicleID, Type, Manufacturer

Subclass: Car

- Attributes: Model, Color

Weak-entity-type representation:



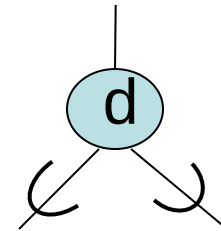
Impact on schema structure, semantics, operations, and design philosophy:

- Is the above representation *equivalent* to the original hierarchical representation?
- Can any subclasses be converted into a weak entity type while preserving data integrity?

Disjointness Constraints

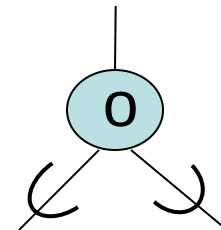
- **Disjoint (d)**

- The subclasses must have **disjoint** sets of entities
- i.e. a secretary *cannot* be in the engineer subclass too



- **Overlap (o)**

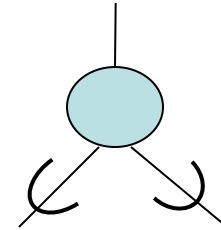
- The subclasses may have **overlapping** sets of entities
- i.e. the same entity may be a member of *more than one* subclass of the specialization



Completeness Constraints

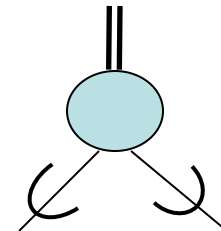
- **Partial**

- An entity **may not** belong to any of the subclasses (single-line)



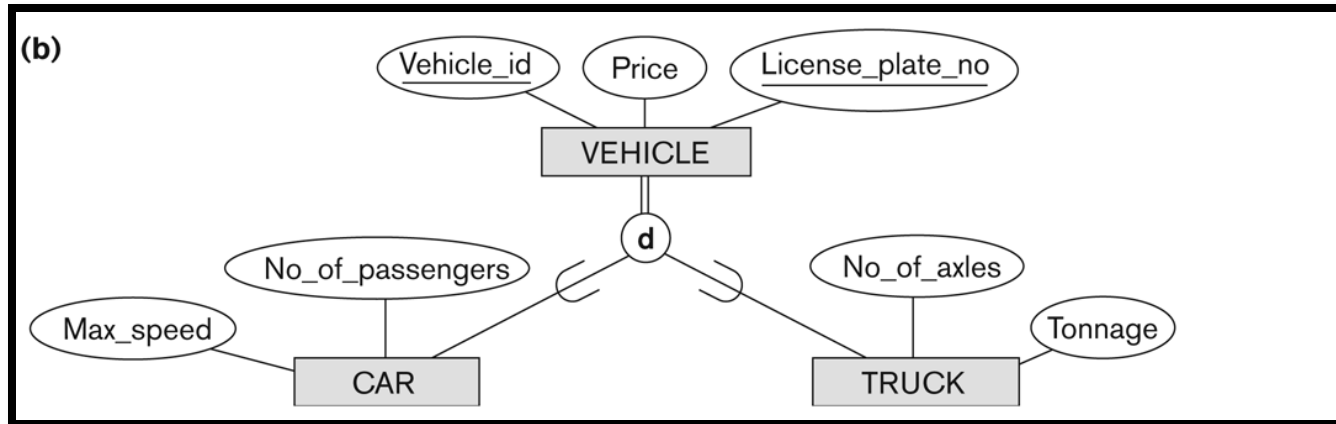
- **Total**

- Every entity in the superclass **must be** a member of some subclass (double-edge)



Question

Can a vehicle in the 'VEHICLE' entity set exist without being a car or a truck?



A) Yes

B) No

Specialization and Generalization Hierarchies and Lattices

- A subclass itself may have further subclasses specified on it, forming a **hierarchy** or a **lattice** of specializations/generalizations
- ***Specialization/Generalization Hierarchy***
 - every subclass has only one superclass
 - ***single inheritance***
 - tree structure or *strict hierarchy*

Specialization and Generalization Hierarchies and Lattices

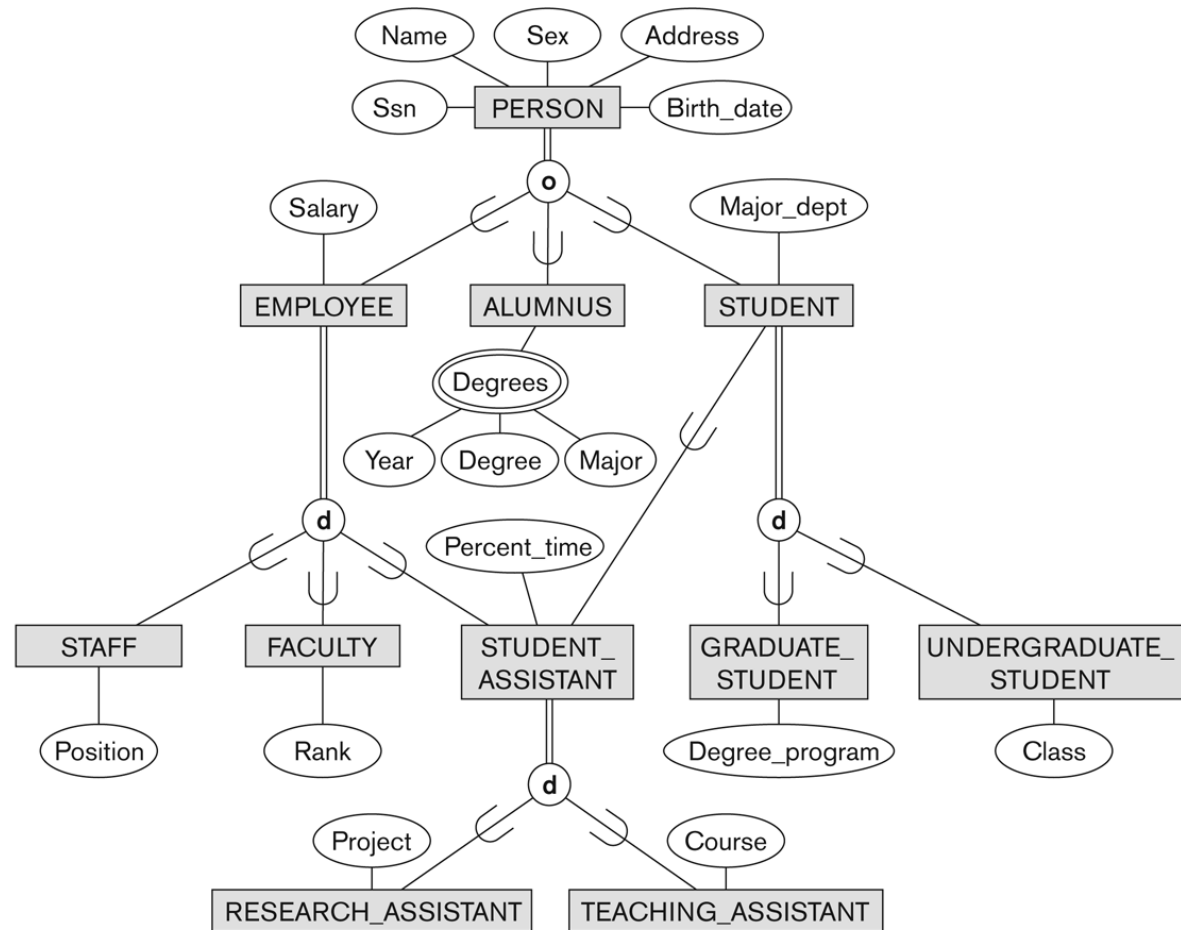


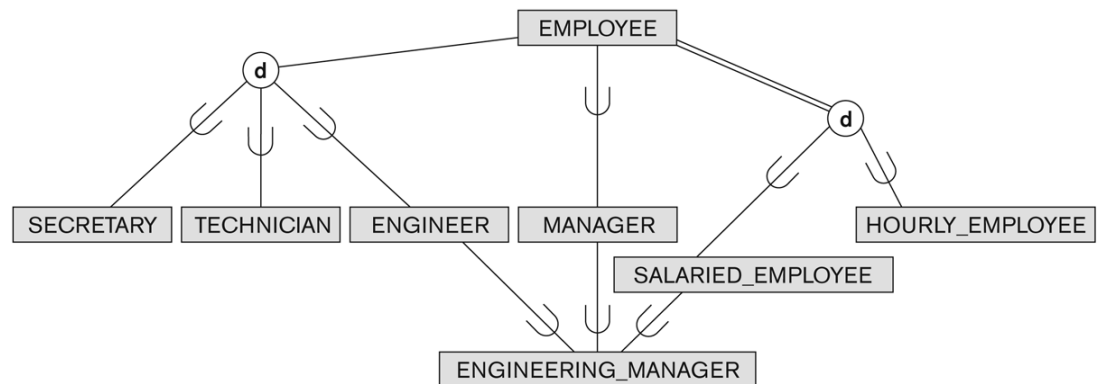
Figure 4.7

A specialization lattice with multiple inheritance for a UNIVERSITY database.

Identify all the *single inheritance* in this EER diagram.

Specialization and Generalization Hierarchies and Lattices

- **Specialization/Generalization Lattice**
 - a subclass can be subclass of more than one superclass
 - **multiple inheritance**
 - a subclass that has **multiple parents**
 - parents of a shared subclass do NOT require a common ancestor
 - also called a **shared subclass**
 - defined to be a subset of the **intersection** of its superclasses
 - a shared subclass member **MUST** exist in **ALL** of its superclasses



Is this diagram showing a specialization hierarchy or a lattice?

Specialization and Generalization Hierarchies and Lattices

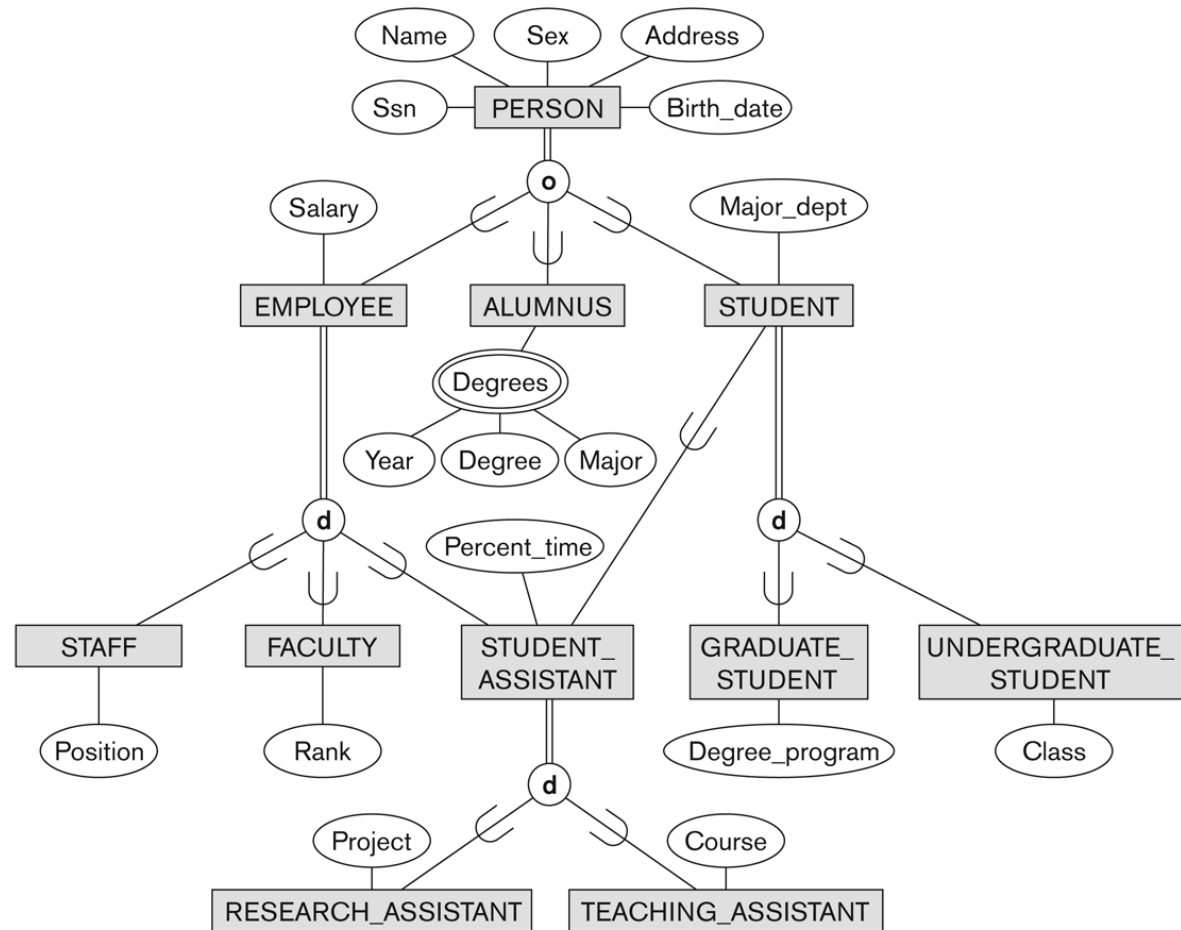
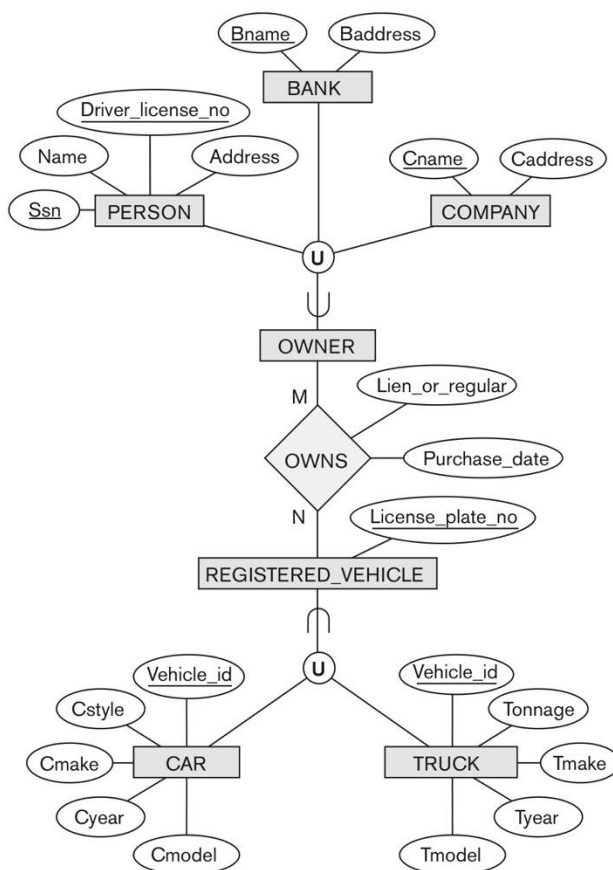


Figure 4.7

A specialization lattice with multiple inheritance for a UNIVERSITY database.

Identify all the shared subclasses in this EER diagram.



- The subclass **OWNER** represents a **union** of *different entity types*
- An entity that is a member of OWNER must exist in *only one* of the superclasses
- An OWNER entity may be a **person**, a **company**, or a **bank**

Modeling of UNION Types Using Categories

- It is sometimes necessary to represent a collection of entities from *different entity types*
- A subclass is often used to represent such a collection of entities
 - union type or a category

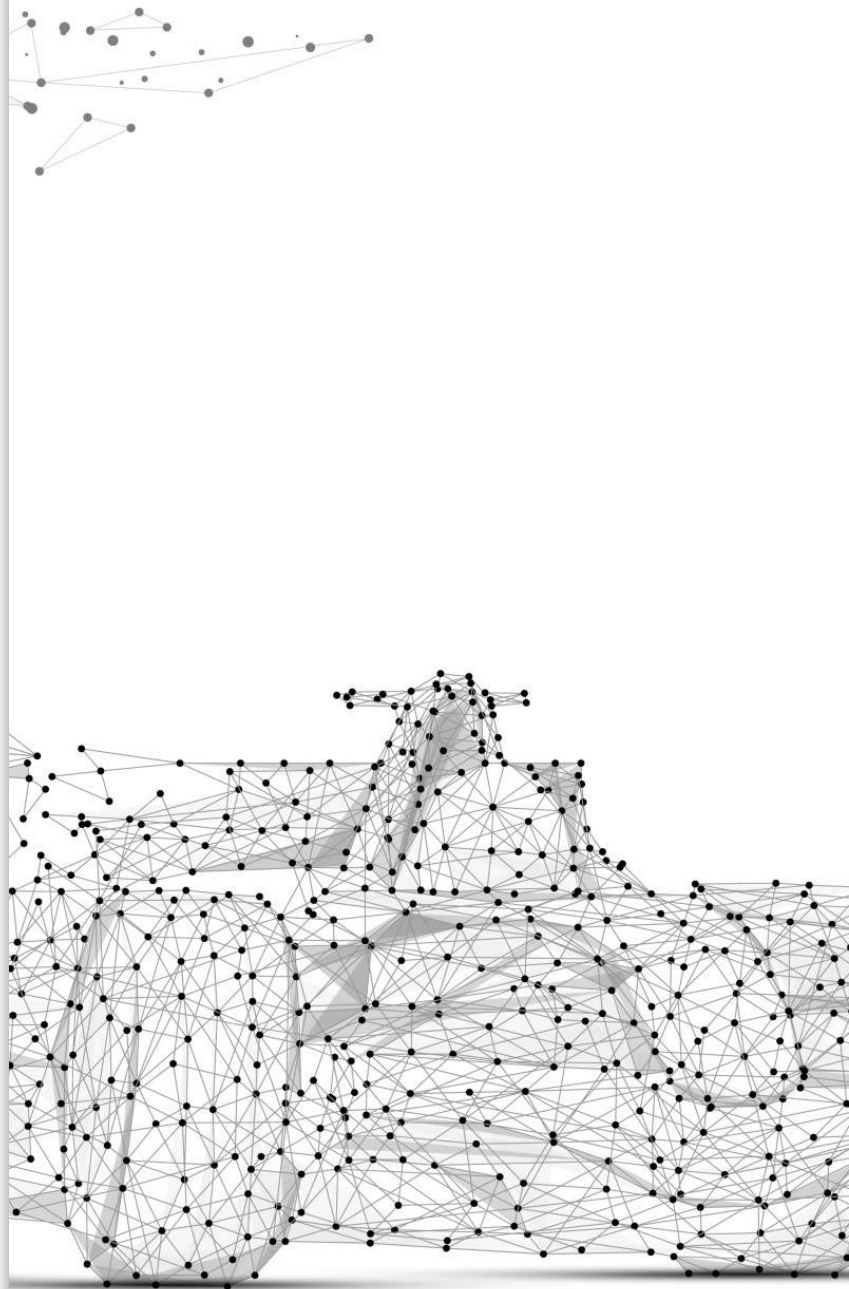
Scenario Exercise

Consider an application with two independent entity types: a **Vehicle** class and a **ComputerizedSystem** class.

The Vehicle entity type has attributes (id, make, model, speed)

The ComputerizedSystem entity type has attributes (id, processing power, software).

Task: Model a new entity type called **SelfDrivingCar** which requires data from both Vehicle and ComputerizedSystem.



Any volunteers? +1 bonus point!

Scenario Exercise

Consider an application with two independent entity types: a **Vehicle** class and a **ComputerizedSystem** class.

The Vehicle entity type has attributes (id, make, model, speed)

The ComputerizedSystem entity type has attributes (id, processing power, software).

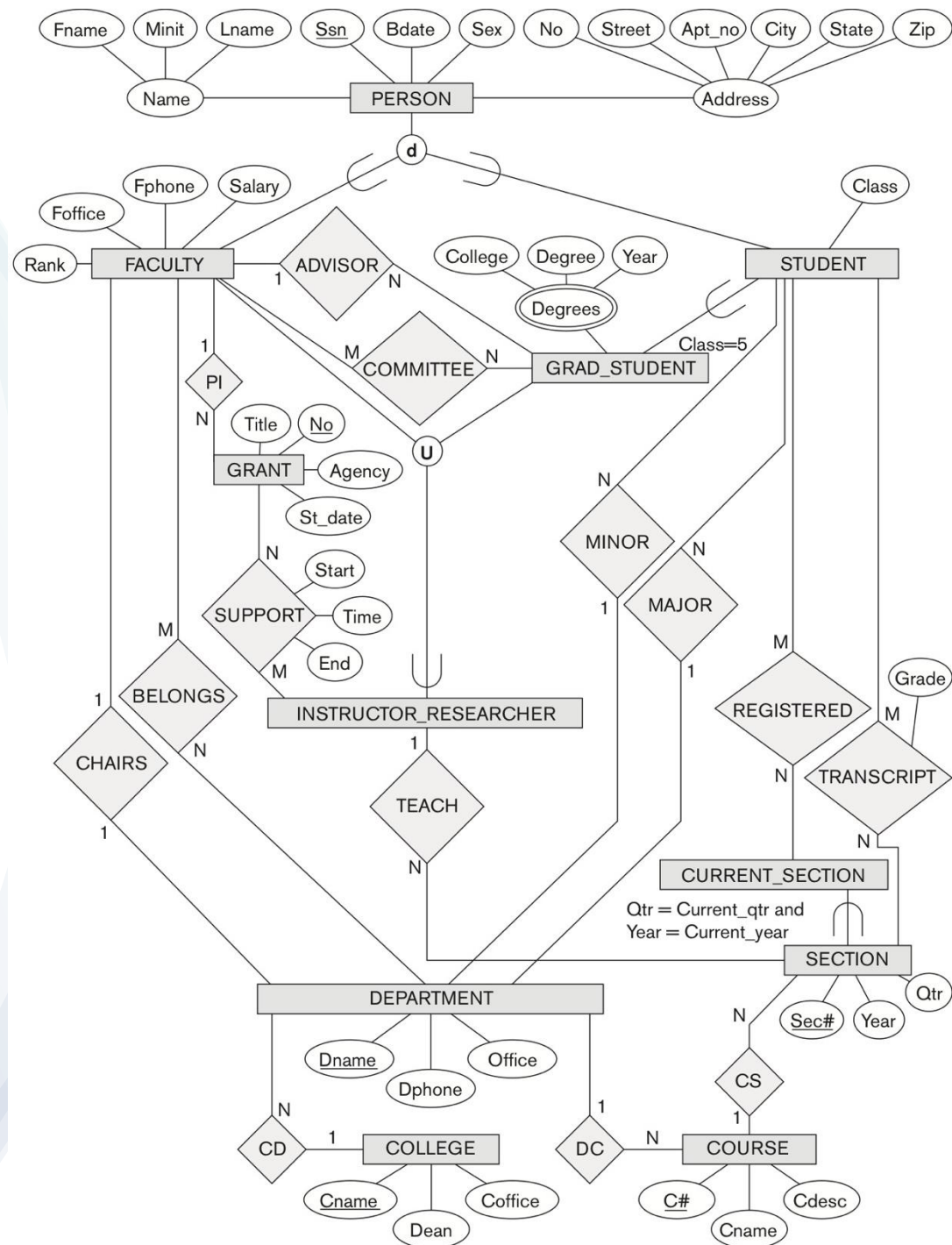
Task: Model a new entity type called **SelfDrivingCar** which requires data from both Vehicle and ComputerizedSystem.

A Sample UNIVERSITY EER Schema

Note:

The subclass **CURRENT_SECTION** (of **SECTION**) is defined based on the following **predicates**:

- *Qtr* = *Current_qtr* (*quarter*)
- *Year* = *current_year*
- each section (key = *Sec#*) is offered at a particular time defined by (*Year*, *Qtr*)





Summary of Lecture 4

The Enhanced Entity-Relationship (EER) Model

- **What we learned?**
 - Subclasses and Superclasses
 - EER Model Inheritance
 - Specialization
 - Generalization
 - Constraints on Specialization and Generalization
 - Hierarchies and Lattices
 - Union Type
- **What you understood the most?**
- **What are the most difficult to you?**